For cover page.

1. Given the stress tensor $\boldsymbol{\sigma}$ in three-dimensional Cartesian coordinates system as

$$\boldsymbol{\sigma} = \begin{bmatrix} 1 & 6 & 0 \\ 6 & -4 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

Please answer the following questions.

(a) (4 points) Calculate the hydrostatic stress component p and the deviatoric stress component $\boldsymbol{s}$ of $\boldsymbol{\sigma}$.

Solution: According to the definition of $p = \sigma_{kk}/3$, we have

$$p = \frac{1 + (-4) + 3}{3} = 0$$

Therefore, the deviatoric stress tensor $\boldsymbol{s}$ is

$$\boldsymbol{s} = \boldsymbol{\sigma} - p\mathbf{1} = \boldsymbol{\sigma} = \begin{bmatrix} 1 & 6 & 0 \\ 6 & -4 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

where $\mathbf{1}$ is the three-dimensional identity tensor

$$\mathbf{1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(b) (6 points) Find the three principal stresses $\sigma_1 \geq \sigma_2 \geq \sigma_3$ for this stress tensor $\boldsymbol{\sigma}$.

Solution: According to the definition of the principal stress σ, which is copied below

$$\begin{vmatrix} \sigma_{11} - \sigma & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} - \sigma & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} - \sigma \end{vmatrix} = 0$$

We would get the following cubic algebraic equation of σ

$$\Big[(1 - \sigma)(-4 - \sigma) - 6^2\Big](3 - \sigma) = 0$$

The three roots of the above equation are easily found as follows

$$\sigma = 3, \quad 5, \quad -8$$

By ordering these three roots, we can get the σ_1, σ_2, and σ_3 as

$$\sigma_1 = 5\,, \quad \sigma_2 = 3\,, \quad \sigma_3 = -8$$

Total for Question 1: 10

2. Please answer the following questions about constitutive model.

(a) (3 points) List three elements of the plastic constitutive relation of soil, no mathematical equations are required.

Solution: The answers are (1 point for each): Yield function (Elastic limit surface); Plastic flow rule; Hardening/softening law.

(b) (3 points) In this course, we have learned the Mohr-Coulomb yield criterion, which can be expressed through the maximum and minimum principal stresses $\sigma_{\max}$ and $\sigma_{\min}$ as

$$(\sigma_{\max} - \sigma_{\min}) + (\sigma_{\max} + \sigma_{\min}) \sin\phi = 2c\cos\phi$$

where ϕ is the friction angle and c is the cohesion. Suppose we know the value of friction angle as $\phi = \pi/6$, please calculate the value of cohesion c such that the stress tensor $\boldsymbol{\sigma}$ in Problem 1 is located exactly on the yield surface.

Solution: From Problem 1, we know $\sigma_{\max} = 5$ and $\sigma_{\min} = -8$. By substituting these values into the Mohr-Coulomb yield criterion, we can solve for c as

$$c = \frac{(\sigma_{\max} - \sigma_{\min}) + (\sigma_{\max} + \sigma_{\min}) \sin\phi}{2\cos\phi} = \frac{13 + (-3) \times 1/2}{2 \times \sqrt{3}/2} = \frac{11.5}{\sqrt{3}} \approx 6.6395$$

Note that the answer 6.64 or 6.6 would also receive the full credit.

(c) (4 points) From the linear elastic Hooke's law which is copied here using index notation

$$\sigma_{ij} = \lambda \epsilon_{kk} \delta_{ij} + 2G\epsilon_{ij}$$

where δ_{ij} is the Kronecker delta symbol, λ and G are Lamé constants. Now suppose we are in the **plane strain** situation, please derive the corresponding two-dimensional elastic bulk modulus K_{2D} using λ and G. The definition of K_{2D} is given as

$$K_{2D} = \frac{(\sigma_{xx} + \sigma_{yy})/2}{\epsilon_{xx} + \epsilon_{yy}}$$

Solution: For plane strain case, we know $\epsilon_{zz} \equiv 0$, and by using Hooke's law, we have

$$\sigma_{xx} = \lambda(\epsilon_{xx} + \epsilon_{yy}) + 2G\epsilon_{xx}\,, \quad \sigma_{yy} = \lambda(\epsilon_{xx} + \epsilon_{yy}) + 2G\epsilon_{yy}$$

By adding the above two equations, we have

$$\sigma_{xx} + \sigma_{yy} = 2(\lambda + G)(\epsilon_{xx} + \epsilon_{yy})$$

which implies

$$K_{2D} = \frac{(\sigma_{xx} + \sigma_{yy})/2}{\epsilon_{xx} + \epsilon_{yy}} = \lambda + G$$

Total for Question 2: 10

3. (20 points) Please evaluate the ultimate bearing capacity P_u of a purely cohesive soil with cohesion c_u, assuming a translational mechanism whose failure surface is sketched in Figure 1. In the figure, all the triangles are isosceles triangles and they are congruent to each other (i.e., $\overline{GF} = \overline{AF} = \overline{AC} = \overline{BC} = \overline{BD} = \overline{DE}$ and $\overline{GA} = \overline{AB} = \overline{BE} = \overline{FC} = \overline{CD}$), Points G, A, B, and E are collinear (line GABE is horizontal), Points F, C, and D are collinear (line FCD is horizontal).

Hint: Note that we have talked about a similar problem with $\beta = \pi/3$ during the lecture. Now we are asking for the results valid for a general $\beta \in (0, \pi/2)$.

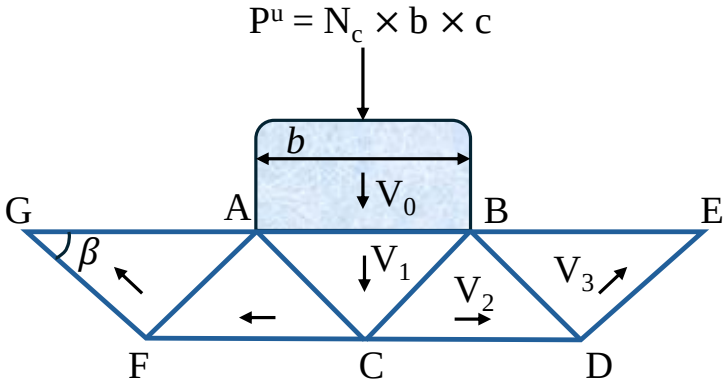


Figure 1: Sketch of one type of translational failure mechanism.

Solution: Since there is no relative motion in vertical direction between the foundation and block $\triangle ABC$, we have $V_1 = V_0$.

Next, by requiring that the relative velocity V_{12} between $\triangle ABC$ and $\triangle BCD$ must align with the line BC and the $\triangle BCD$ would move horizontally, we could construct the velocity vectorial triangle and solve for V_2 and V_{12} as

$$V_2 = \frac{V_1}{\tan\beta}, \quad V_{12} = \frac{V_1}{\sin\beta}$$

Similarly, by requiring that the relative velocity V_{23} between $\triangle BCD$ and $\triangle BDE$ must align with the line BD and the $\triangle BDE$ would move along the line DE, we could solve for V_3 and V_{23} as

$$V_3 = V_{23} = \frac{V_2}{2\cos\beta} = \frac{V_1}{2\sin\beta}$$

Now we need to extract some geometry information. According to the description, we know all the isosceles triangles have a base length of b and a leg length of $b/(2\cos\beta)$.

Finally, we could write out the expression for the rate of external virtual work δW and the rate of internal energy dissipation δD as

$$\begin{aligned}
\delta W &= P_u V_0 = P_u V_1 \\
\delta D &= 2c_\mathrm{u}\left(\overline{BC}\times V_{12} + \overline{CD}\times V_2 + \overline{BD}\times V_{23} + \overline{DE}\times V_3\right) \\
&= 2c_\mathrm{u}\left[\frac{b}{2\cos\beta}\times\left(V_{12}+V_{23}+V_3\right) + b\times V_2\right] \\
&= 2c_\mathrm{u}\left[\frac{b}{2\cos\beta}\times\frac{2V_1}{\sin\beta} + b\times\frac{V_1}{\tan\beta}\right] \\
&= 2bc_\mathrm{u}V_1\left[\frac{1}{\sin\beta\cos\beta} + \frac{1}{\tan\beta}\right]
\end{aligned}$$

By equating δW with δD, we get

$$P_u V_1 = 2bc_{\mathrm{u}} V_1 \left[\frac{1}{\sin\beta\cos\beta} + \frac{1}{\tan\beta} \right]$$

which gives

$$\frac{P_u}{bc_{\mathrm{u}}} = \underbrace{2\left(\frac{1}{\sin\beta\cos\beta} + \frac{1}{\tan\beta} \right)}_{N_c}$$

Total for Question 3: 20

4. Please answer the following questions about slope stability analysis

   (a) (5 points) In LEM-based method of slices, based on the following Figure 2 which depicts the forces acted on one slice, please state the assumptions on **interslice forces** for (1) Ordinary method and (2) Bishop's simplified method.

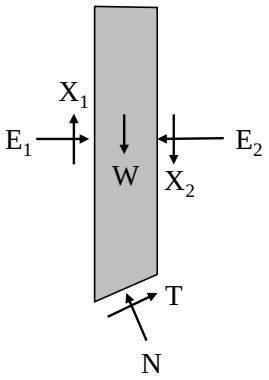


Figure 2: Forces acted on one slice in LEM-based method of slices

> **Solution:** For the Ordinary method, the rigorous version of the answer is: the resultants of E_1 and X_1 are equal in magnitude to the resultants of E_2 and X_2 and their lines of action coincide. However, the answers such as $E_1 = E_2 = 0$ and $X_1 = X_2 = 0$ are also acceptable.
>
> For Bishop's simplified method, the assumption is that **ONLY** X_1 and X_2 would cancel with each other. There is no assumption on the horizontal interslice forces E_1 and E_2.

   (b) (15 points) Given coordinates O(11,20) [m], F(14,14) [m] and soil properties: $c' = 10$ kPa, $\phi' = 35°$, $\gamma = 11$ kN/m^3 (dry condition, no pore water pressure). For the slice from AA' ($x_A = x_{A'} = 22$ [m]) to BB' ($x_B = x_{B'} = 24$ [m]), which is shown in the following Figure 3, please evaluate the **normal force** N and **shear force** T mobilized at the base of this slice by assuming a safety factor of $F_s = 1.5$ in the Bishop's simplified method.

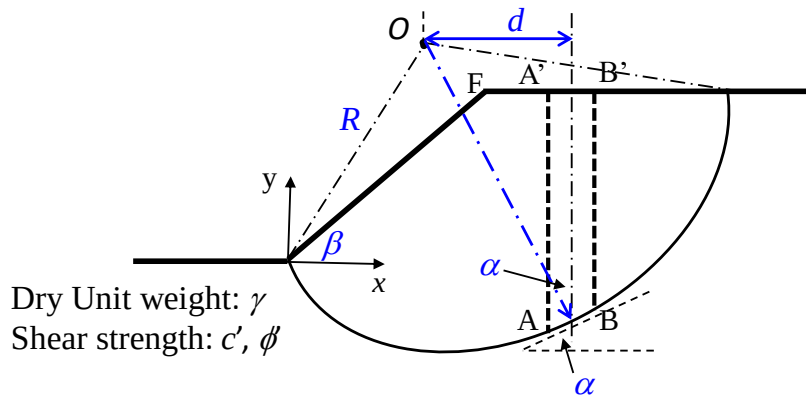


Figure 3: Slope stability calculation for one slice using the Bishop's simplified method.

Solution: First of all, we would need to calculate geometric information of this slice such as the slice height H. The radius of the circular failure surface is $R = \sqrt{11^2 + 20^2} = \sqrt{521}$ [m], and the coordinates y_A and y_B are (in [m])

$$y_A = y_O - \sqrt{R^2 - (x_A - x_O)^2} = 20 - \sqrt{521 - (22 - 11)^2} = 0$$

$$y_B = y_O - \sqrt{R^2 - (x_B - x_O)^2} = 20 - \sqrt{521 - (24 - 11)^2} = 20 - 4\sqrt{22} \approx 1.2383$$

Therefore, the slice height H is calculate as (in [m])

$$H = y_F - \frac{y_A + y_B}{2} = 4 + 2\sqrt{22} \approx 13.38$$

and the slice width is just the difference between x_A and x_B, which is 2 [m]. As a result, the slice weight is (assuming unit thickness 1 [m] in z direction)

$$W = \gamma \times b \times H = 44(2 + \sqrt{22}) \approx 294.3783\,\text{kN}$$

We also need to calculate the angle α in order to evaluate the length $L = \overline{AB} = 2/\cos\alpha$ (in [m])

$$\alpha = \arcsin \frac{(x_A + x_B)/2 - x_O}{R} = \arcsin \frac{12}{\sqrt{521}} \approx 0.5536$$

$$L = \frac{x_B - x_A}{\cos\alpha} \approx 2.3511$$

Finally, we resort to the vertical force equilibrium and we have

$$W = N\cos\alpha + T\sin\alpha$$

By using the definition of safety factor F_s, we also have

$$T = \frac{N\tan\phi' + c'L}{F_s}$$

A combination of the previous two equations yields our desired result for N as well as T since W, α, F_s, c', and ϕ' are given (two linear equations with two unknowns). The solution reads (in [kN])

$$N = 261.0592\,, \quad T = 137.5380$$

Specifically, they are solved through this equation

$$\begin{bmatrix} N \\ T \end{bmatrix} = \begin{bmatrix} \cos\alpha & \sin\alpha \\ -\dfrac{\tan\phi'}{F_s} & 1 \end{bmatrix}^{-1} \begin{bmatrix} W \\ \dfrac{c'L}{F_s} \end{bmatrix}$$

Total for Question 4: 20

5. As shown in Figure 4, a two-bar system is subjected to an external force F on the right. Each bar element has a Young's modulus of E_1 and E_2, cross-sectional areas of A_1 and A_2, and lengths of l_1 and l_2, respectively.

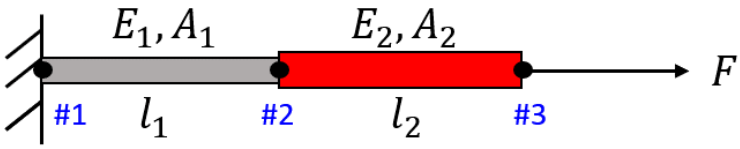


Figure 4: Two bar system subjected to external load

(a) (4 points) Please directly present stiffness equation for the system in the matrix form.

Solution: The relationship between the nodal forces and displacements can be represented in matrix form as $\mathbf{Ku} = \mathbf{F}$, where:

The displacement vector is:

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$$

The force vector is:

$$\mathbf{F} = \begin{bmatrix} F_1 \\ F_2 \\ F_3 \end{bmatrix}$$

The global stiffness matrix $\mathbf{K}$ for the system is:

$$\mathbf{K} = \begin{bmatrix} \frac{E_1A_1}{l_1} & -\frac{E_1A_1}{l_1} & 0 \\ -\frac{E_1A_1}{l_1} & \frac{E_1A_1}{l_1} + \frac{E_2A_2}{l_2} & -\frac{E_2A_2}{l_2} \\ 0 & -\frac{E_2A_2}{l_2} & \frac{E_2A_2}{l_2} \end{bmatrix}$$

(b) (4 points) Suppose two bars are identical, i.e., $E_1 = E_2 = E$, $A_1 = A_2 = A$, $l_1 = l_2 = l$, please compute the displacement for node #3

Solution: The simplified matrix equation is given as:

$$\frac{EA}{l} \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} 0 \\ F \end{bmatrix}$$

We can express this as two separate equations:

$$\frac{EA}{l}(2u_2 - u_3) = 0$$
$$\frac{EA}{l}(-u_2 + u_3) = F$$

Thus, it is easy to have:

$$u_3 = \frac{2Fl}{EA}, u_2 = \frac{Fl}{EA}$$

Total for Question 5: 8

6. We have a rectangular block with the following geometry and mesh discretization as shown in Figure 5. The thickness of the problem is set to 1 unit. The lower left corner of the block is fixed, and its upper right corner is subjected to a nodal force of $F = 10$ N. The elastic parameters are as follows: Young's modulus $E = 30$ Pa, Poisson's ratio $\nu = 1/3$, and the plane stress condition is assumed.

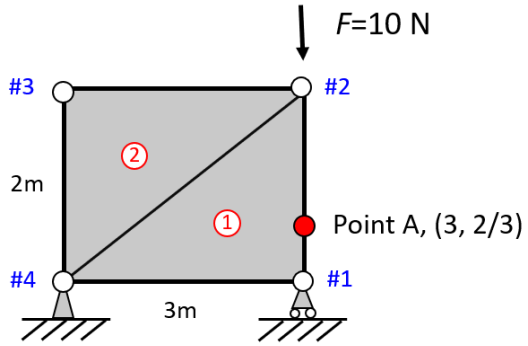


Figure 5: Forces acted on rectangular block

(a) (10 points) Please briefly describe the principle of the finite element method (FEM) and list the computational procedure for the FEM;

Solution:
Principle: Discretize the whole domain into simple elements, making the complex, intractable problem into a set of simpler, solvable problems.
Computational procedure:

1. Establish the finite element model.
2. Compute the elastic stiffness matrix.
3. Compute the strain–displacement matrix.
4. Compute the element stiffness matrix.
5. Assemble the global stiffness matrix.
6. Impose boundary conditions.
7. Solve the system of equations.
8. Post-analysis (evaluate strain, stress, and contour plots).

Note: Steps 2 and 3 are interchangeable and step 8 is optional.

(b) (4 points) Please compute the shape function N_I of Point A (I = 1 to 4)

Solution: $N_1 = 2/3$, $N_2 = 1/3$, $N_3 = 0$ and $N_4 = 0$,

(c) (6 points) Suppose the node order is [1, 2, 4] in element 1 and [3, 4, 2] in element 2. Now, we have computed the element stiffness matrix K_e for both elements as below. Please compute the k_{13}, k_{22}, k_{34}, and k_{77} values in the assembled global stiffness matrix K.

$$K_e^1 = K_e^2 = \begin{bmatrix} 19.6875 & -11.2500 & -8.4375 & 5.6250 & -11.2500 & 5.6250 \\ -11.2500 & 29.0625 & 5.6250 & -25.3125 & 5.6250 & -3.7500 \\ -8.4375 & 5.6250 & 8.4375 & 0 & 0 & -5.6250 \\ 5.6250 & -25.3125 & 0 & 25.3125 & -5.6250 & 0 \\ -11.2500 & 5.6250 & 0 & -5.6250 & 11.2500 & 0 \\ 5.6250 & -3.7500 & -5.6250 & 0 & 0 & 3.7500 \end{bmatrix}$$

$$K_{global} = \begin{bmatrix} * & . & k_{13} & . & . & . & . & . \\ & k_{22} & . & . & . & . & . & . \\ & & * & k_{34} & . & . & . & . \\ & & & * & . & . & . & . \\ & & & & * & . & . & . \\ & & & & & * & . & . \\ & \text{sym.} & & & & & k_{77} & . \\ & & & & & & & * \end{bmatrix}$$

Solution: For K_e^1, its DOF mapping is

	DOF 1	DOF 2	DOF 3	DOF 4	DOF 7	DOF 8
DOF 1	19.6875	−11.2500	−8.4375	5.6250	−11.2500	5.6250
DOF 2	−11.2500	29.0625	5.6250	−25.3125	5.6250	−3.7500
DOF 3	−8.4375	5.6250	8.4375	0	0	−5.6250
DOF 4	5.6250	−25.3125	0	25.3125	−5.6250	0
DOF 7	−11.2500	5.6250	0	−5.6250	11.2500	0
DOF 8	5.6250	−3.7500	−5.6250	0	0	3.7500

For K_e^2, its DOF mapping is

	DOF 5	DOF 6	DOF 7	DOF 8	DOF 3	DOF 4
DOF 5	19.6875	−11.2500	−8.4375	5.6250	−11.2500	5.6250
DOF 6	−11.2500	29.0625	5.6250	−25.3125	5.6250	−3.7500
DOF 7	−8.4375	5.6250	8.4375	0	0	−5.6250
DOF 8	5.6250	−25.3125	0	25.3125	−5.6250	0
DOF 3	−11.2500	5.6250	0	−5.6250	11.2500	0
DOF 4	5.6250	−3.7500	−5.6250	0	0	3.7500

then we can compute k_{13}, k_{22}, k_{34}, and k_{77} as below:

$$\begin{aligned} k_{13} &= (-8.4375) + (0) &&= -8.4375 \\ k_{22} &= (29.0625) + (0) &&= 29.0625 \\ k_{34} &= (0) + (0) &&= 0 \\ k_{77} &= (11.2500) + (8.4375) &&= 19.6875 \end{aligned}$$

Total for Question 6: 20

7. Consider the one-dimensional heat conduction equation without sources, which is a second-order time-dependent partial differential equation (PDE):

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2},$$

where $u(x,t)$ represents the temperature at position x and time t, and α is the thermal diffusivity constant. Let Δx represent the spatial step size and Δt represent the time step size. The temperature at a grid point in space x_i and time t^n is represented by u_i^n.

(a) (3 points) Approximate the second-order spatial derivative $\dfrac{\partial^2 u}{\partial x^2}$ using a central difference scheme.

Solution: The second-order spatial derivative can be approximated using a central difference scheme as follows:

$$\frac{\partial^2 u}{\partial x^2} \approx \frac{u(x+\Delta x,t) - 2u(x,t) + u(x-\Delta x,t)}{(\Delta x)^2}.$$

(b) (3 points) Please discretize the above PDE with FTCS (Forward Time, Centered Space) and BTCS (Backward Time, Centered Space) schemes.

Solution:
The FTCS scheme can be written as:

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} = \alpha \frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{(\Delta x)^2}$$

Rearranging for u_i^{n+1}, we obtain:

$$u_i^{n+1} = u_i^n + \alpha \frac{\Delta t}{(\Delta x)^2}(u_{i+1}^n - 2u_i^n + u_{i-1}^n)$$

The BTCS scheme can be formulated as:

$$\frac{u_i^n - u_i^{n-1}}{\Delta t} = \alpha \frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{(\Delta x)^2}$$

and can be rewritten as

$$-\alpha \frac{\Delta t}{(\Delta x)^2} u_{i-1}^n + \left(1 + 2\alpha \frac{\Delta t}{(\Delta x)^2}\right) u_i^n - \alpha \frac{\Delta t}{(\Delta x)^2} u_{i+1}^n = u_i^{n-1}$$

This leads to a system of equations that must be solved for u_i^n at each time step.

Total for Question 7: 6

8. (6 points) Refer to Figure 6. Two identical disks, each with a mass of $m = 10\,\text{kg}$ and a radius of $r = 0.5\,\text{m}$, are in contact. The disks interact through an elastic contact described by a linear contact law with a stiffness of $k = 1\,\text{kN/m}$. The centroids of the disks are located at $x_1 = 0.5\,\text{m}$ and $x_2 = 1.4\,\text{m}$, respectively.

   Assuming that at $t = 0\,\text{s}$, both disks have initial velocities of $v = 0\,\text{m/s}$, and there are no external forces applied, calculate the updated velocities of both disks at $t = 2 \times 10^{-5}\,\text{s}$ using a Discrete Element Method (DEM) simulation with a time step of $\Delta t = 10^{-5}\,\text{s}$.

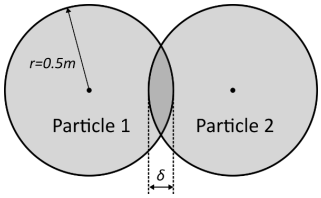


Figure 6: Geometry and initial position of two bodies in contact

Solution: Given:

- Spring constant $k = 1000\,\text{N/m}$
- Initial compression $\delta_0 = 0.1\,\text{m}$
- Disk mass $m = 10\,\text{kg}$
- Time step $\Delta t = 10^{-5}\,\text{s}$

The initial force is calculated as:

$$F_0 = k\delta_0 = 1000 \times 0.1 = 100\,\text{N}$$

The initial acceleration is:

$$a_0 = \frac{F_0}{m} = \frac{100}{10} = 10\,\text{m/s}^2$$

The displacement after the first time step is:

$$d = \frac{1}{2}a_0\Delta t^2 = \frac{1}{2} \times 10 \times (10^{-5})^2 = 5 \times 10^{-10}\,\text{m}$$

The overlap after the first time step is:

$$\delta_{\Delta t} = \delta_0 - 2d \approx 0.1\,\text{m} - 1 \times 10^{-9}\,\text{m} \approx 0.1\,\text{m}$$

The force after the first time step (assuming negligible change):

$$F_{\Delta t} \approx F_0 = 100\,\text{N}$$

The velocity after the first time step is:

$$v_{\Delta t} = a_0\Delta t = 10 \times 10^{-5} = 1 \times 10^{-4}\,\text{m/s}$$

The updated velocity after the second time step is:

$$v_{2\Delta t} = v_{\Delta t} + a_0\Delta t = 1 \times 10^{-4} + 1 \times 10^{-4} = 2 \times 10^{-4}\,\text{m/s}$$

It is also possible to compute the accurate contact force in the second step, they are all correct.

Total for Question 8: 6

Question	Points	Score
1	10	
2	10	
3	20	
4	20	
5	8	
6	20	
7	6	
8	6	
Total:	100	

- End of Paper -